

EXTREME NETWORKS

Apply Config Template XIQ-SE workflow

Ludovico Stevens

Technical Marketing Engineering

July 2022

Apply Config Template XIQ-SE workflow



Workflow to apply an ASCII config template containing variables. Works with any device family

The embedded variables can be of four types:

- `${variableName}`: XIQ-SE Global or Site specific variables, in this preference order: local site, site parents, global
 - Selected `emc_vars` can be included by adding them to `const_EXPORT EMC_VARS`; currently:
 - `"deviceIP","deviceName","deviceSysOid","deviceType","family","serverIP","serverName","date"`
- `$<csvColumnKey>`: Device specific variables extracted from supplied CSV file
- `$(variableName)`: Variables set from within the config template using `#eval variableName=()`
- `$UD1, $UD2, $UD3, $UD4`: Device specific values extracted from device User Data 1-4

For the CSV variables, a CSV file must be provided with the following syntax:

- First row has column labels, which need to match the `$<csvColumnKey>` variables, without the `$<>`
- Subsequent rows contain data values for every device, one row per device.
- First column contains the device lookup, either the device IP or Serial Number

ASCII config template file and the CSV file must be placed on the XIQ-SE filesystem.

Workflow manual execution



- Workflow can be manually run against 1 or many switches simultaneously

Devices Sandbox ZTF Site Summary Endpoint Locations FlexReports

+ Add Device... Export to CSV

Status	Name ↑	Site	IP Address	Poll Status	Poll Details	Device Type	Family
▲	Sbox-VSP7200-1	/World/PoC/Sandb...	10.8.4.2	Available: 1...	Up: 412 Do...	VSP-7254XSQ	VSP Series
▲	Sbox-VSP7200-2	/Wo		Available: 1...	Up: 412 Do...	VSP-7254XSQ	VSP Series
▲	Sbox-VSP7200-3	/Wo		Available: 1...	Up: 411 Do...	VSP-7254XSQ	VSP Series
▲	Sbox-VSP7200-4	/Wo		Available: 1...	Up: 412 Do...	VSP-7254XSQ	VSP Series
●	X460G2-1	/Wo		Available: 1...	Up: 412 Do...	X460-G2-24t-10G4	Summit Se...

FlexView
More Views
Configure...
Compass Search...
Rediscover
Clear Alarms...
Upgrade Firmware...
Add to Device Group...
More Actions
Archives
Tasks
Maps
Network
Policy

Access Control
Config
Example
Macro
Provisioning

Apply Config Template
CLI Custom Action - XOS SSL
Delete Insight VMs
Deploy Insight VM
Fabric Attach Enforce
Fabric Connect Enforce
Move to CLIP Mgmt IP
SMLT Pair Enforce
by Marlon - Onboard VSP

Workflow automatic execution during onboarding



- Workflow can be automatically run after ZTP+ onboarding, under XIQ-SE Site Actions
- In this case script will always run against 1 switch only, the onboarding switch

Devices **Sandbox** Site Summary Endpoint Locations FlexReports

Discover **Actions** VRF/VLAN Topologies Services Port Templates ZTP+ Device Defaults Endpoint Locations Analytics Custom Variables

☒ Automatically Add Devices Collection Mode: Historical

☒ Add Trap Receiver Collection Interval (minutes): 10

☒ Add Syslog Receiver

☒ Add to Archive

☐ Add to Map

Custom Configuration

+ Add ✎ Edit ✖ Delete

Enabled	Vendor	Family	Topology	Task
☒	Extreme	Universal Platform S	Any	Provisioning/Apply Config Template

Update Cancel

Workflow inputs, with CSV values



Run Workflow - Apply Config Template

Workflow Inputs

Custom Inputs

Notes:

Inputs below can be set either with absolute values or can be provided as \${<site-custom-variable>} if the workflow is to derive a site specific value for those inputs. The site of the device, and parent sites, will apply.

ASCII config template:

/root/Ludo/template.cfg

CSV data file:

/root/Ludo/variable-values.csv

Index into CSV file:

IP address

IP address

Serial Number

Sanity:

Debug:

*template.cfg - Notepad

File Edit Format View Help

```
link-state group 1 enable
link-state group 1 upstream interface gigabitEthernet $UD1
link-state group 1 downstream interface gigabitEthernet ${tuni port}
router vrf fabext
  ip route 0.0.0.0 0.0.0.0 $UD2 weight 10
  ip route ${tunnelDest} 255.255.255.255 $UD2 weight 10
exit
router isis
  ip-tunnel-source-address ${deviceIP} vrf fabext
  manual-area ${isis area}
exit
logical-intf isis 255 dest-ip ${tunnelDest} name ${tunnel name}
  isis
  isis spbm 1
  isis spbm 1 ll-metric ${vxlan nni metric}
  isis enable
exit
filter acl 1 type inPort
filter acl port 1 ${tuni port}
filter acl ace 1 1001
filter acl ace action 1 1001 permit remark-dot1p ${macsec qos remark}
filter acl ace action 1 1001 permit count
filter acl ace action 1 1001 permit internal-qos ${macsec qos remark}
filter acl ace ethernet 1 1001 dst-mac mask 00:00:00:00:00:00 0xffffffffffff
filter acl ace 1 1001 enable
i-sid ${tuni isid} elan-transparent
  port ${tuni port}
exit
```

variable-values.csv													
File Home Insert Draw Page Layout Formulas Data Review View Automate Help													
A1	ip												
	A	B	C	D	E	F	G	H	I	J	K	L	M
1	ip	mask	macsec nni	macsec nni metric	tuni port	tunnel dest	isis area	tunnel name	vxlan nni metric	macsec qos remark	tuni isid	macsec key	macsec parity
2	10.8.4.2	255.255.255.0	1/21	20	1/22	10.10.10.10	47	DC1	10		4	15000001	1234567890 odd
3	10.8.4.3	255.255.255.0	1/21	20	1/22	10.10.10.20	47	DC2	10		4	15000002	1234567890 odd
4	10.8.4.4	255.255.255.0	1/21	20	1/22	10.10.10.10	47	DC1	10		4	15000003	1234567890 odd
5	10.8.4.5	255.255.255.0	1/21	20	1/22	10.10.10.20	47	DC2	10		4	15000004	1234567890 odd

- If your XIQ-SE was installed without “root” access, place the CSV file here instead:
/usr/local/Extreme_Networks/NetSight/appdata/logs/scripting/NetSight_Server

Workflow inputs, with CSV values



Run Workflow - Apply Config Template

Workflow Inputs

Custom Inputs

Notes:
Inputs below can be set either with absolute values or can be provided as `$(site-custom-variable)` if the workflow is to derive a site specific value for those inputs. The site of the device, and parent sites, will apply. Or alternatively as `$` to derive a device specific value extracted from the supplied CSV file. For template and CSV file the following path variables are available: `%rootDir%`, `%sitePath%`, `%siteName%`

ASCII config template:
Fabric Engine edge

CSV data file:
/root/mgmtdata.csv

Index into CSV file:
Serial Number

Skip if device not found in CSV file:
disable

Device notes:
Configured by Apply Config Template workflow on `$(date)`

Follow on workflow notes:

Configuration Templates

Templates

/Device Type (3 templates)

Name	Template Name	Archive Name	Archive Version	Archive De Type
Device Type (3 templates)				
Avaya (Rapid City) (1 template)	Fabric Engine edge			
Extreme (1 template)	Core VSP8600			
All Templates (3 templates)	ISW			

Edit Configuration Template - Fabric Engine edge

```
1  
2 config terminal  
3  
4 clock time-zone Europe London  
5  
6 ip domain-name "reading.ctc.local"  
7 ip name-server primary 10.8.255.160  
8 ip name-server secondary 10.8.255.159  
9
```

variable-values.csv

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	ip	mask	macsec nni	macsec nni metric	tuni port	tunnel dest	isis area	tunnel name	vlan nni metric	macsec qos remark	tuni isid	macsec key	macsec parity
2	10.8.4.2	255.255.255.0	1/21	20	1/22	10.10.10.10	47	DC1	10	4	15000001	1234567890	odd
3	10.8.4.3	255.255.255.0	1/21	20	1/22	10.10.10.20	47	DC2	10	4	15000002	1234567890	odd
4	10.8.4.4	255.255.255.0	1/21	20	1/22	10.10.10.10	47	DC1	10	4	15000003	1234567890	odd
5	10.8.4.5	255.255.255.0	1/21	20	1/22	10.10.10.20	47	DC2	10	4	15000004	1234567890	odd

- The config template can be a configuration template defined in XIQ-SE; in this case provide the template name with no path or file extension

Workflow inputs, with CSV values



Run Workflow - Apply Config Template

Workflow Inputs

Custom Inputs

Notes:

Inputs below can be set either with absolute values or can be provided as `${<site-custom-variable>}` if the workflow is to derive a site specific value for those inputs. The site of the device, and parent sites, will apply.

ASCII config template:

/root/\${<template>}

CSV data file:

/root/Ludo/variable-values.csv

Index into CSV file:

IP address

IP address

Serial Number

are not actually made. Debug: enable if you need to repo

Sanity:

Debug:

variable-values.csv														
File Home Insert Draw Page Layout Formulas Data Review View Automate Help														
N1 : X ✓ fx template														
	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	ip	mask	macsec nni	macsec nni metric	tuni port	tunnel dest	isis area	tunnel name	vxlan nni metric	macsec qos remark	tuni isid	macsec key	macsec parity	template
2	10.8.4.2	255.255.255.0	1/21	20	1/22	10.10.10.10	47 DC1		10	4	15000001	1234567890	odd	template.cfg
3	10.8.4.3	255.255.255.0	1/21	20	1/22	10.10.10.20	47 DC2		10	4	15000002	1234567890	odd	template.cfg
4	10.8.4.4	255.255.255.0	1/21	20	1/22	10.10.10.10	47 DC1		10	4	15000003	1234567890	odd	template2.cfg
5	10.8.4.5	255.255.255.0	1/21	20	1/22	10.10.10.20	47 DC2		10	4	15000004	1234567890	odd	template2.cfg

*template.cfg - Notepad

File Edit Format View Help

```
link-state group 1 enable
link-state group 1 upstream interface gigabitEthernet $UD1
link-state group 1 downstream interface gigabitEthernet ${<tuni port>}
router vrf fabext
  ip route 0.0.0.0 0.0.0.0 $UD2 weight 10
  ip route ${<tunnelDest>} 255.255.255.255 $UD2 weight 10
exit
router isis
  ip-tunnel-source-address ${<deviceIP>} vrf fabext
  manual-area ${<isis area>}
exit
logical-intf isis 255 dest-ip ${<tunnelDest>} name ${<tunnel name>}
  isis
  isis spbm 1
  isis spbm 1 ll-metric ${<vxlan nni metric>}
  isis enable
exit
filter acl 1 type inPort
filter acl port 1 ${<tuni port>}
filter acl ace 1 1001
filter acl ace action 1 1001 permit remark-dot1p ${<macsec qos remark>}
filter acl ace action 1 1001 permit count
filter acl ace action 1 1001 permit internal-qos ${<macsec qos remark>}
filter acl ace ethernet 1 1001 dst-mac mask 00:00:00:00:00:00 0xffffffffffff
filter acl ace 1 1001 enable
i-sid ${<tuni isid>} elan-transparent
  port ${<tuni port>}
exit
```

- Note that the CSV can also be used to specify the template file to use (if you have more than 1 template)

Workflow, path variables



Run Workflow - Apply Config Template

Workflow Inputs

Custom Inputs

Notes:

Inputs below can be set either with absolute values or can be provided as `${<site-custom-variable>}` if the workflow is to derive a site specific value for those inputs. The site of the device, and parent sites, will apply. For template and csv file the following path variables are available: `%rootDir%`, `%sitePath%`, `%siteName%`.

ASCII config template:

`%rootDir%/%sitePath%/%siteName%.cfg`

CSV data file:

`%rootDir%/%sitePath%/%siteName%.csv`

Index into CSV file:

Serial Number

IP address

Serial Number

MAC Address

Sanity: enable if you do not trust this workflow and wish to first see what it does. In sanity mode...

*template.cfg - Notepad

```
File Edit Format View Help
link-state group 1 enable
link-state group 1 upstream interface gigabitEthernet $UD1
link-state group 1 downstream interface gigabitEthernet ${tuni port}
router vrf fabext
  ip route 0.0.0.0 0.0.0.0 $UD2 weight 10
  ip route ${tunnelDest} 255.255.255.255 $UD2 weight 10
exit
router isis
  ip-tunnel-source-address ${deviceIP} vrf fabext
  manual-area ${isis area}
exit
logical-intf isis 255 dest-ip ${tunnelDest} name ${tunnel name}
isis
  isis spbm 1
  isis spbm 1 ll-metric ${vxlan nni metric}
  isis enable
exit
filter acl 1 type inPort
filter acl port 1 ${tuni port}
filter acl ace 1 1001
filter acl ace action 1 1001 permit remark-dot1p ${macsec qos remark}
filter acl ace action 1 1001 permit count
filter acl ace action 1 1001 permit internal-qos ${macsec qos remark}
filter acl ace ethernet 1 1001 dst-mac mask 00:00:00:00:00:00 0xffffffffffff
filter acl ace 1 1001 enable
i-sid ${tuni isid} elan-transparent
  port ${tuni port}
exit
```

variable-values.csv

File Home Insert Draw Page Layout Formulas Data Review View Help

A1 : X ✓ f x ip

- Available path variables: **%rootDir%**, **%sitePath%**, **%siteName%**
 - `%rootDir%` by default is `/root/`; can be changed via workflow variable `const_ROOT_PATH_VAR`
 - `%sitePath%` and `%siteName%` are set based on site path of device; e.g. if device is in `"/World/CTC-Reading/VSP Sandbox"` then `%sitePath%` = `"World/CTC-Reading"` and `%siteName%` = `"VSP Sandbox"`
- Can use these to have different CSV & template files per site

Workflow, site variables

- Template can also take \${var} variables
- Values for these variables are looked up in the Site Custom variables, in this preference order:
 - Site of device
 - Parent Site of device
 - Parent sites up to Root site
 - Global variable

```
*template.cfg - Notepad
File Edit Format View Help
link-state group 1 enable
link-state group 1 upstream interface gigabitEthernet $UD1
link-state group 1 downstream interface gigabitEthernet $<tuni port>
router vrf fabext
  ip route 0.0.0.0 0.0.0.0 $UD2 weight 10
  ip route ${tunnelDest} 255.255.255.255 $UD2 weight 10
exit
router isis
  ip-tunnel-source-address ${deviceIP} vrf fabext
  manual-area $<isis area>
exit
logical-intf isis 255 dest-ip ${tunnelDest} name $<tunnel name>
  isis
  isis spbm 1
  isis spbm 1 ll-metric $<vxlan nni metric>
  isis enable
exit
filter acl 1 type inPort
filter acl port 1 $<tuni port>
filter acl ace 1 1001
filter acl ace action 1 1001 permit remark-dot1p $<macsec qos remark>
filter acl ace action 1 1001 permit count
filter acl ace action 1 1001 permit internal-qos $<macsec qos remark>
filter acl ace ethernet 1 1001 dst-mac mask 00:00:00:00:00:00 0xffffffffffff
  ace 1 1001 enable
  ni isid> elan-transparent
  tuni port>
```

Devices Sandbox ZTF Site Summary Endpoint Locations FlexReports					
Discover Actions VRF/VLAN Topologies Services Port Templates ZTP+ Device Defaults Endpoint Locations Analytics Custom Variables					
Add Edit Delete					
Scope		Variable			
Category	Site ↓	Type	Name	Type	Value
Site	/World/PoC/Sandbox ZTF		locationGroup	String	PhysicalSandbox
Site	/World/PoC/Sandbox ZTF		dataIsid	String	0
Site	/World/PoC/Sandbox ZTF		tunnelDest	String	20.120.10.20
Site	/World/PoC/Sandbox ZTF		dataVlan	String	0
Site	Global		Auto Sync VLANs in...	String	Done
Site	Global		nacEnable	String	enable
Site	Global		locationGroup	String	0

Workflow, emc_vars as variables

- Selected emc_vars can also be fed into the same \${var} space, by adding them to workflow variable const_EXPORT_EMCC_VARS

The screenshot shows a workflow editor interface. The main workspace displays a workflow diagram with a 'Start' node, an 'Apply Config Template' task, and an end node labeled 'Signal - Config Applied'. The 'Variables' tab is active, showing a table of workflow variables.

Name	Default Value
const_CSV_DELIMITER	,
const_EXPORT_EMCC_VARS	["deviceIP", "deviceName", "deviceSysOid", "deviceType", "family", "ports", "serverIP", "serverName"]
devices	
input_configFile	
input_csvFile	
input_csvKey	

```
*template.cfg - Notepad
File Edit Format View Help
link-state group 1 enable
link-state group 1 upstream interface gigabitEthernet $UD1
link-state group 1 downstream interface gigabitEthernet $<tuni port>
router vrf fabext
  ip route 0.0.0.0 0.0.0.0 $UD2 weight 10
  ip route ${tunnelDest} 255.255.255.255 $UD2 weight 10
exit
router isis
  ip-tunnel-source-address ${deviceIP} vrf fabext
  manual-area $<isis area>
exit
logical-intf isis 255 dest-ip ${tunnelDest} name $<tunnel name>
  isis
  isis spbm 1
  isis spbm 1 ll-metric $<vxlan nni metric>
  isis enable
exit
filter acl 1 type inPort
filter acl port 1 $<tuni port>
filter acl ace 1 1001
filter acl ace action 1 1001 permit remark-dot1p $<macsec qos remark>
filter acl ace action 1 1001 permit count
filter acl ace action 1 1001 permit internal-qos $<macsec qos remark>
filter acl ace ethernet 1 1001 dst-mac mask 00:00:00:00:00:00 0xffffffffffff
filter acl ace 1 1001 enable
i-sid $<tuni isid> elan-transparent
  port $<tuni port>
exit
```

Workflow, UserData1-4 variables

- Values unique to each device can also be fetched from the device UserData1-4 fields
- Only value after “=” is used

The screenshot shows the 'Configure Device' interface. On the left, a table lists devices with columns for Status, Name, and a dropdown menu. The selected device is 'Sbox-VSP7200-2'. The main form contains fields for Device ID, System Name, Device Nickname, and Device Type. Below these are tabs for Device, Device Annotation, VRF Definitions, VLAN Definitions, CLIP Addresses, Topology, and Services. The 'Device Annotation' tab is active, showing fields for Nickname, Asset Tag, and four User Data fields. The 'User Data 1' field contains 'wan port = 1/1' and the 'User Data 2' field contains 'wan router = 192.168.255.123'. Both fields are highlighted with red boxes.

Device ID	System Name	Device Nickname	Device Type
10.8.4.3	Sbox-VSP7200-2	Sbox-VSP7200-2	VSP-7254XSQ

Device: Sbox-VSP7200-2

Asset Tag:

User Data 1: wan port = 1/1

User Data 2: wan router = 192.168.255.123

User Data 3:

User Data 4:

Note:

```
*template.cfg - Notepad
File Edit Format View Help
link-state group 1 enable
link-state group 1 upstream interface gigabitEthernet $UD1
link-state group 1 downstream interface gigabitEthernet ${tuni port}
router vrf fabext
  ip route 0.0.0.0 0.0.0.0 $UD2 weight 10
  ip route ${tunnelDest} 255.255.255.255 $UD2 weight 10
exit
router isis
  ip-tunnel-source-address ${deviceIP} vrf fabext
  manual-area ${isis area}
exit
logical-intf isis 255 dest-ip ${tunnelDest} name ${tunnel name}
  isis
  isis spbm 1
  isis spbm 1 ll-metric ${vxlan nni metric}
  isis enable
exit
filter acl 1 type inPort
filter acl port 1 ${tuni port}
filter acl ace 1 1001
filter acl ace action 1 1001 permit remark-dot1p ${macsec qos remark}
filter acl ace action 1 1001 permit count
filter acl ace action 1 1001 permit internal-qos ${macsec qos remark}
filter acl ace ethernet 1 1001 dst-mac mask 00:00:00:00:00:00 0xffffffffffff
filter acl ace 1 1001 enable
i-sid ${tuni isid} elan-transparent
  port ${tuni port}
exit
```

Workflow, Template eval variables



File Edit View

```
#if ($<nodeType> == "core")
    #eval locArea=('30.0'+$<locId>)
    #eval remArea=('30.0000')
    #eval systemId=('020c.0' + $<locId> + '.0' + $<nodeId> + '0')
    #eval nickName=($<locId>[0] + '.' + $<locId>[1:3] + '.' + $<nodeId>)
    #eval vnSystemId=('9200.' + $<locId> + '0.FFF0')
    #eval vnNick=('9.a'+$<locId>[0] + '.' + $<locId>[1:3])
    #eval remSysId=('040c.0' + str(int($<locId>)+300) + '.0' + $<nodeId> + '0')
    #eval remNick=(str(int($<locId>)+300)[0] + '.' + str(int($<locId>)+300)[1:3] + '.' + $<nodeId>)
    #eval remVnSysId=('940c.0' + str(int($<locId>)+300) + '.FFF0')
    #eval vnNick=('9.b'+$<locId>[0] + '.' + $<locId>[1:3])
    #if (int($<nodeId>) % 2)
        #eval virtualBmac = ('02:0d:0' + $<locId>[0] + ':' + $<locId>[1:3] + ':0' + $<nodeId>[0] + '0')
        #eval peerSysId = ('020d.0' + $<locId> + '.00' + str(int($<nodeId>)+1) + '0')
        #eval peerIp = (int($<nodeId>)+1)
    #else
        #eval virtualBmac = ('02:0c:0' + $<locId>[0] + ':' + $<locId>[1:3] + ':0' + format(str(int($<nodeId>)+1), '0'))
        #eval peerSysId = ('020d.0' + $<locId> + '.00' + str(int($<nodeId>)-1) + '0')
        #eval peerIp = (int($<nodeId>)-1)
    #end
    config term
    #block start
        no router isis enable\ny
        router isis
        manual-area $[locArea]
        system-id $[systemId]
        spbm 1 nick-name $[nickName]
        area-vnode sys-name #eval ('vnode-'+$[locArea])
    exit
    router isis remote
        manual-area 49.00bb
        system-id $[remSysId]
        spbm 1 nick-name $[remNick]
        area-vnode sys-name vnode-49.00bb
    exit
    router isis remote enable
    router isis enable
#block execute 5
#end
```

	A	B	C	D	E	F	G
1	Serial Number	mgmtClip	sysName	siteName	locId	nodeId	nodeType
2	1902Q-20003	10.10.0.3	DCA-408	/World/DCA	010	03	core
3	1902Q-20011	10.10.0.4	DCA-508	/World/DCB	010	04	core
4	SB012105G-00040	10.15.0.1	CMPa-421	/World/CMPa	015	01	disti
5	SB012106G-00149	10.15.0.2	CMPa-422	/World/CMPa	015	02	disti
6	JA142141G-00126	10.15.0.11	CMPa-424	/World/CMPa	015	11	access
7	JA142143G-00044	10.15.0.12	CMPa-425	/World/CMPa	015	12	access
8	SB012106G-00058	10.16.0.1	CMPb-521	/World/CMPb	016	01	disti
9	SB012106G-00026	10.16.0.2	CMPb-522	/World/CMPb	016	02	disti
10	JA122133G-00608	10.16.0.11	CMPb-524	/World/CMPb	016	11	access
11	JA072336G-00100	10.16.0.12	CMPb-525	/World/CMPb	016	12	access

- Template `$(var)` variables are set within the config template script itself, via use of the `#eval` pragma
- These are handy for computing values using `#eval`, and storing the result if it needs to be re-used several times (or just to make things neater)

Comparison of template variables



- Site variables **`${var}`**: Useful to apply same values to all devices in same XIQ-SE Site. Or to apply same values to all devices in same sub-Sites
- CSV variables **`$<var>`**: Useful to provide device specific values
- UserData variables **`$UDI-4`**: Useful to provide device specific values, but for values obtained dynamically from the device itself (by another upstream workflow or activity) and then make these available in this Apply Config Template workflow
- **`Emc_vars`** `${deviceIP}`: Useful to feed some of these values into the same space as Site variables
- Template eval variables **`$(var)`**: Useful to compute new values within the template file and be able to store and re-use these values via a variable

CSV “peer:” values



- In the example shown, assume that the workflow is run against switch with S/N 2128Q-40044
- Variable `$<sysID>` will thus resolve to 02bb.7400.0200
- Cell A1 in the CSV, normally is just a label of the lookup index to the CSV values (serial number here). But it can be augmented with “:<other CSV variable>” (“:smltVirtBmac” here)
- When the CSV is parsed it will now not only return all values for the lookup serial number (2128Q-40044) but will also inspect the values of smltVirtBmac and if another entry in the CSV has the same value for smltVirtBmac then all CSV variables for that other entry will also become available in the config template as `$<peer:var>`. Hence `$<peer:sysID>` will resolve to 02bb.7400.0300
- The expectation is that only 2 entries in the CSV file will have the same value for the selected smltVirtBmac column; if more than 2 are found CSV parsing will halt with an error

```
act-template.txt
File Edit View

#block start
no router isis enable\ny
router isis
  system-id $<sysID>
  spbm 1 smlt-peer-system-id $<peer:sysID>
  spbm 1 smlt-virtual-bmac $<smltVirtBmac>
exit
vlan create 4000 name "IST-VLAN" type port-mstprstp 0
vlan i-sid 4000 $<ist isid>
interface Vlan 4000
  #if(int($<sysID>.replace(".", ""), 16) < int($<peer:sysID>.replace(".", ""), 16))
  ip address 192.168.255.1 255.255.255.252
  exit
  virtual-ist peer-ip 192.168.255.2 vlan 4000
  #else
  ip address 192.168.255.2 255.255.255.252
  exit
  virtual-ist peer-ip 192.168.255.1 vlan 4000
#end
router isis enable
#block execute 60
```

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
	serial number	mgmt vlanid	mgmt isid	mgmt ip	mgmt clip	mgmt mask	mgmt gateway	sysname	site name	comments	next workflow	sysID	smltVirtBmac	ist isid
1	2128Q-40044	129	2102901	10.129.0.72		255.255.255.0	10.129.0.1	Sbox-VSP7400-2		Kevin	Provisioning/Apply Config Template	02bb.7400.0200	02:bb:74:00:02:ff	15000001
2	2128Q-40009	129	2102901	10.129.0.73		255.255.255.0	10.129.0.1	Sbox-VSP7400-3		Stuart	Provisioning/Apply Config Template	02bb.7400.0300	02:bb:74:00:02:ff	15000001
3	1916Q-20031	129	2102901	10.129.0.74		255.255.255.0	10.129.0.1	Sbox-VSP7400-4		Gru	Provisioning/Apply Config Template	02bb.7400.0400		

Commands which need to be sourced locally on the switch



- There may be situations where a bunch of commands need to be executed directly by the switch itself, like for configuration that would otherwise make the switch temporarily unreachable to XIQ-SE, e.g. shutting down ISIS, in order to change the fabric IDs
 - The commands are packed into a text file, which is TFTP downloaded to the switch and executed locally using the source command
 - On successful completion the file is removed from the switch, and the TFTP boot flag is disabled, if it was disabled to start with
- Of course this is risky; the commands need to be validated to ensure that XIQ-SE connectivity will be restored at the end of the block sequence
- Note this functionality will only work with VOSS/FabricEngine, EXOS/SwitchEngine and BOSS/ERS
- Note that this will not work with VOSS/FabricEngine “no ssh”, as that command will kill the SSH session right away and nothing will get sourced locally
- Other velocity type statements (#if/#elseif/#else/#end/#error..) cannot be used inside the #block statements; but variables can be used
- Syntax (NOTE: these are case sensitive pragmas, use lower case “block”, “start”, “execute” only and no space between # and command):

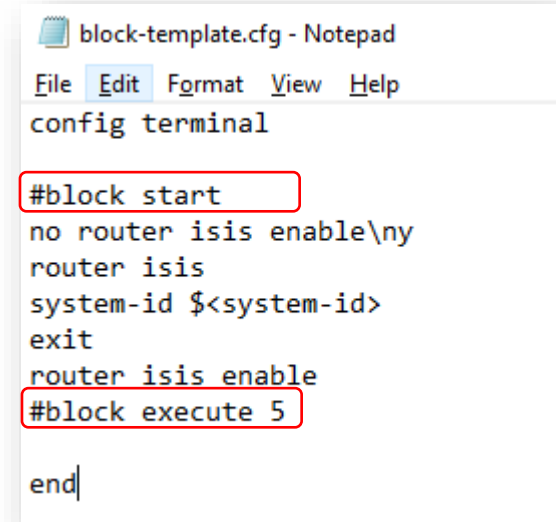
#block start [n] : Mark the beginning of a block of commands which will need to be sourced locally on the switch.

[n] = Optional number of seconds to sleep after block execution

#block execute [wait|reconnect] [n]: Mark the end of block of commands which are to sourced locally on the switch. If this directive is not seen, and the “#block start” was seen, all commands from the start of block section to the last command in the template file will be sourced locally on the switch.

[wait|reconnect] = “wait”, after waiting [n] secs the same CLI session is resumed; while “reconnect”, after still waiting [n] secs, will tear down the CLI session and start a new CLI session; use this mode if [n] is getting > 90sec, as the switch might end up timing out the CLI session; “wait” is the default mode if none is specified

[n] = Optional number of seconds to sleep after block execution



```
block-template.cfg - Notepad
File Edit Format View Help
config terminal

#block start
no router isis enable\ny
router isis
system-id $<system-id>
exit
router isis enable
#block execute 5

end
```


Cisco velocity type statements: #if/#elseif/#else/#end



- The template file can also include **#if/#elseif/#else/#end** statement blocks
- To match Cisco velocity type statements
- No space between # and command
- The conditional string, inside “(“”)” will be evaluated using Python’s eval() function, so any valid Python expression may be used
- Any of CSV variables \$<var>, Site variables \${var}, UserData variables \$UD1-4 or template variables \$[var] can be inserted inside the conditional string, but they will always be evaluated as String values.
- For instance, to evaluate an integer value, insert the variable inside Python’s int() method like this:
 - `#if ( int($<myvar>) > 10)`
- NOTE: these are case sensitive pragmas, use lower case (if,elseif,else,end) only

```
template2.cfg - Notepad
File Edit Format View Help
config terminal
interface GigabitEthernet $<myport>
#if($<myport> == "1/1")
    name "first port"
#elif($<myport> == "1/24")
    name "middle port"
#else
    name "last port"
#end
exit
```

```
template3.cfg
File Edit View
config terminal
#if("5520-24" in ${deviceType})
interface gigabitEthernet 1/1-1/24
    no snmp trap link-status
exit
#elif("5520-48" in ${deviceType})
interface gigabitEthernet 1/1-1/48
    no snmp trap link-status
exit
#end
```


Cisco velocity type statements: #if/#elseif/#else/#end - nesting



File Edit View

```
#if ($<nodeType> == "core")
    #eval locArea=('30.0'+$<locId>)
    #eval remArea=('30.0000')
    #eval systemId=('020c.0' + $<locId> + '.0' + $<nodeId> + '0')
    #eval nickName=($<locId>[0] + '.' + $<locId>[1:3] + '.' + $<nodeId>)
    #eval vnSystemId=('9200.' + $<locId> + '0.FFF0')
    #eval vnNick=('9.a'+$<locId>[0] + '.' + $<locId>[1:3])
    remSysId=('040c.0' + str(int($<locId>)+300) + '.0' + $<nodeId> + '0')
    remNick=(str(int($<locId>)+300)[0] + '.' + str(int($<locId>)+300)[1:3] + '.' + $<nodeId>)
    remVnSysId=('940c.0' + str(int($<locId>)+300) + '.FFF0')
    #eval vnNick=('9.b'+$<locId>[0] + '.' + $<locId>[1:3])
    #if (int($<nodeId>) % 2)
        #eval virtualBmac = ('02:0d:0' + $<locId>[0] + ':' + $<locId>[1:3] + ':0' + $<nodeId>)
        #eval peerSysId = ('020d.0' + $<locId> + '.00' + str(int($<nodeId>)+1) + '0')
        #eval peerIp = (int($<nodeId>)+1)
    #else
        #eval virtualBmac = ('02:0c:0' + $<locId>[0] + ':' + $<locId>[1:3] + ':0' + format(str(int($<nodeId>)-1), '02x'))
        #eval peerSysId = ('020d.0' + $<locId> + '.00' + str(int($<nodeId>)-1) + '0')
        #eval peerIp = (int($<nodeId>)-1)
    #end
    config term
    #block start
        no router isis enable\nv
        router isis
        manual-area $[locArea
        system-id $[systemId
        spbm 1 nick-name $[nickName
        area-vnode sys-name
        exit
        router isis remote
        manual-area 49.00bb
        system-id $[remSysId
        spbm 1 nick-name $[remNick
        area-vnode sys-name
        exit
        router isis remote enable
        router isis enable
    #block execute 5
#end
```

act-template.txt

File Edit View

```
#block start
no router isis enable\nv
router isis
    system-id $<sysID>
    spbm 1 smlt-peer-system-id $<peer:sysID>
    spbm 1 smlt-virtual-bmac 02:bb:ff:#eval(format(int($<cluster-id>), '02x')):00:ff
exit
vlan create 4000 name "IST-VLAN" type port-mstprstp 0
vlan i-sid 4000 $<ist isid>
interface Vlan 4000
    #if(int($<sysID>.replace(".", ""), 16) < int($<peer:sysID>.replace(".", ""), 16))
        ip address 192.168.255.1 255.255.255.252
        exit
        virtual-ist peer-ip 192.168.255.2 vlan 4000
    #else
        ip address 192.168.255.2 255.255.255.252
        exit
        virtual-ist peer-ip 192.168.255.1 vlan 4000
    #end
router isis enable
#block execute 60
```

• **#if/#elseif/#else/#end** statement blocks can be nested, without any limit

• **#if/#elseif/#else/#end** statement blocks can also be used inside **#block start** & **#block execute** sections, as long as they are fully contained inside

Cosmetics only (nesting will work regardless), so that the config applied will look nicer: use tab characters to apply indentation to **#if/#elseif/#else/#end** statement blocks and use spaces for switch command indentation; the final config pushed will suppress the former and keep the latter

Error mode: #error fail|stop|continue



- The template file can also include **#error fail|stop|continue** statement
- No space between # and command
- Determines the behaviour if a command in the template errors when executed on the switch
 - **fail**: workflow aborts immediately with an error
 - **stop**: no further commands from the template are executed, the workflow continues and does not fail
 - **continue**: execution of template commands continues even if commands error
- The default behaviour is **fail**
- NOTE: these are case sensitive pragmas, use lower case only

```
template.cfg - Notepad
File Edit Format View Help
config terminal
link-state group 1 enable
link-state group 1 upstream interface gigabitEthernet $UD1
link-state group 1 downstream interface gigabitEthernet $<tuni port>
#error continue
macsec connectivity-association macsec-profile connectivity-association-key $<macsec key> key-parity $<macsec parity>
#error stop
interface GigabitEthernet $<macsec nni>
    macsec connectivity-association macsec-profile
    macsec encryption enable
    macsec enable
exit
#error fail
filter acl 1 type inPort
filter acl port 1 $<tuni port>
filter acl ace 1 1001
filter acl ace action 1 1001 permit remark-dot1p $<macsec qos remark>
filter acl ace action 1 1001 permit count
filter acl ace action 1 1001 permit internal-qos $<macsec qos remark>
filter acl ace ethernet 1 1001 dst-mac mask 00:00:00:00:00:00 0xffffffffffff
filter acl ace 1 1001 enable
end
Ln 22, Col 1 100% Windows (CRLF) UTF-8
```

Embedded eval: #eval



- Any config line can contain an eval statement:
#eval ()
- No space between # and command
- The string, inside “(“ ”)” will be evaluated using Python’s eval() function and the result converted to string (str)
- In the example the value of \$<cluster-id> is converted to a 2-digit hex number
- NOTE: this is a case sensitive pragma, use lower case “eval” only

```
act-template.txt
File Edit View

#block start
no router isis enable\ny
router isis
  system-id $<sysID>
  spbm 1 smlt-peer-system-id $<peer:sysID>
  spbm 1 smlt-virtual-bmac 02:bb:ff:#eval(format(int($<cluster-id>), '02x')):00:ff
exit
vlan create 4000 name "IST-VLAN" type port-mstprstp 0
vlan i-sid 4000 $<ist isid>
interface Vlan 4000
#if(int($<sysID>.replace(".", ""), 16) < int($<peer:sysID>.replace(".", ""), 16))
  ip address 192.168.255.1 255.255.255.252
  exit
  virtual-ist peer-ip 192.168.255.2 vlan 4000
#else
  ip address 192.168.255.2 255.255.255.252
  exit
  virtual-ist peer-ip 192.168.255.1 vlan 4000
#end
router isis enable
#block execute 60
```

Template eval variables: #eval <varname>=()



- Uses an entire line in the template file:
#eval <varname>=()
- No space between # and command
- This syntax, unlike embedded eval from previous slide, cannot be embedded in commands to be sent to the switch
- The string, inside “(“”)” will be evaluated using Python’s eval() function and the result converted to string (str)
- In the example, fabric ids are evaluated as template variables, then used in commands further down in the template
- NOTE: this is a case sensitive pragma, use lower case “eval” only

```
File Edit View
#eval locArea=('30.0'+$<locId>)
#eval systemId=("020c.0{:03}.0{:02}0".format($<locId>, $<nodeId>))
#eval nickName=($<locId>[0] + '.' + $<locId>[1:3] + '.' + $<nodeId>)
config term
#block start
    no router isis enable\ny
    router isis
        manual-area ${locArea}
        system-id ${systemId}
        spbm 1 nick-name ${nickName}
    exit
    router isis enable
#block execute 5
```

Sleep statement: #sleep



- Introduce a delay while sourcing the template file to the switch:
#sleep <seconds>
- No space between # and command
- Some CLI commands will error if executed too rapidly after the previous command; e.g. on VOSS you have to wait several seconds before enabling RADIUS accounting after having enabled RADIUS...
- The **#sleep** statement cannot be used inside a **#block start** & **#block execute** section
- The <seconds> value can also be supplied by a CSV variable, site variable, template variable, etc..
- NOTE: this is a case sensitive pragma, use lower case "sleep" only

A screenshot of a CLI editor window with a menu bar (File, Edit, View) and a text area containing configuration commands. The commands are: config term, radius enable, #sleep 10, radius accounting enable, and end. The line #sleep 10 is highlighted with a red rectangular box.

```
File Edit View

config term
radius enable
#sleep 10
radius accounting enable
end
```

Last statement: #last



- Do not wait for a prompt to come back on the very last command in the template:
#last
- For example, if the template is to reboot the switch once done
 - Otherwise, if the last command in the template does not receive a prompt, the workflow will timeout and report a failure
- No space between # and command
- The **#last** statement must always appear on the penultimate non-empty line of the template file
 - If it is placed on any other line it will simply be ignored
- Will also work with **#block start** & **#block execute** sections, but must again be placed in line immediately preceding **#block execute** or preceding last line of block section if trailing **#block execute** is omitted
- NOTE: this is a case sensitive pragma, use lower case "last" only

File Edit View

```
config term
boot config flags vrf-scaling
#last
reset -y
```

File Edit View

```
#eval locArea=('30.0'+$<locId>)
#eval systemId=("020c.0{:03}.0{:02}0".format($<locId>, $<nodeId>))
#eval nickName=($<locId>[0] + '.' + $<locId>[1:3] + '.' + $<nodeId>)
config term
#block start
    no router isis enable\ny
    router isis
        manual-area $[locArea]
        system-id $[systemId]
        spbm 1 nick-name $[nickName]
    exit
    router isis enable
    save config
    reset -y
    #last
#block execute 5
```

Commands which require Y/N confirmation prompt on device



- Some commands require “y” confirmation on certain devices (VOSS/ERS)
- To push such commands via the template, append “\ny” to those commands, as shown
- On EXOS the workflow will automatically “disable cli prompting” so this is not necessary
- On VOSS some commands (like “reset”) offer a “-y” switch which can bypass the y/n confirmation prompt; use that if available, else use “\ny”

A screenshot of a Notepad window titled 'template.txt - Notepad'. The window contains a network configuration template for a Cisco-style CLI. The text is as follows:

```
File Edit Format View Help
config term
interface gigabitEthernet 1/1-1/48
    no spanning-tree mstp\ny
exit
end
|
```

The text '\ny' is highlighted in yellow.

Commands which prompt for interactive input on device



- Some commands which set passwords, prompt the user interactively to enter such passwords, sometimes twice
- To push such commands via the template, append the data (passwords) in the same sequence as they would be requested, separated by "//"
- The following example shows how this can be done for "config" which then prompts for:
 - Configuring from terminal or network [terminal]?
To which, "term" will be fed
- The example also shows how to create an SNMPv3 privAuth user which will ask for both the priv password and auth password twice each

A screenshot of a Notepad window titled "template.txt - Notepad". The window contains the following text:

```
File Edit Format View Help
config // term
snmp-server user snmpuser1 group "snmprw" sha aes // snmpauthcred // snmpauthcred // snmpprivcred // snmpprivcred
end
```

Workflow execution



Workflow Dashboard Scheduled Tasks Saved Tasks Scripts Workflows Apply Config Template (26110) ✕

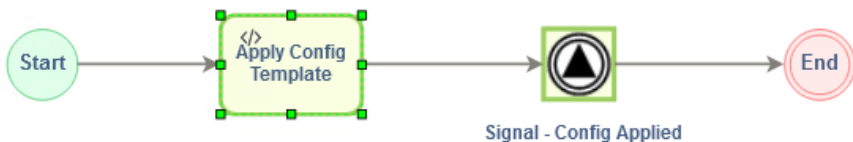
Summary

Status	Start Date/Time	Name	Version	Source	# Devices	Started By	End Date/Time	Message	Path
✓	7/18/2022 2:11:04 ...	Apply Config Templ...	32	Workflow Designer ...	4	Istevens	7/18/2022 2:11:09 ...	Applied Config Template on devices 10.8.4.5...	/Workflows/Ludovico/Apply Config Template

Graph View Table View



Stop Workflow Show Output Show Variables



Devices Grid

Show Output

Status	Device IP	Output Path	Start Date/Time	End Date/Time	Message
SUCCESS	10.8.4.5		7/18/2022 2:1...	7/18/2022 2:1...	Applied Co...
SUCCESS	10.8.4.3		7/18/2022 2:1...	7/18/2022 2:1...	Applied Co...
SUCCESS	10.8.4.4		7/18/2022 2:1...	7/18/2022 2:1...	Applied Co...
SUCCESS	10.8.4.2		7/18/2022 2:1...	7/18/2022 2:1...	Applied Co...

Workflow execution



Workflow Dashboard Scheduled Tasks Saved Tasks Scripts V

Summary

Status	Start Date/Time	Name	Version	So
✓	7/18/2022 2:11:04 ...	Apply Config Templ...	32	Wi

Graph View Table View

🔍 🔍 📖

```
graph LR; Start((Start)) --> Apply[</> Apply Config Template]; Apply --> Signal[Signal - C];
```

Output - 10.8.4.5

```
Script Name: Apply Config Template_Apply_Config_Template
Date and Time: 2022-07-18T14:11:04.670
XIQ-SE User: lstevens
XIQ-SE User Domain:
IP: 10.8.4.5
Workflow version 32 on XIQ-SE/XMC version 22.6.10.70
Activity: ApplyConfigTemplate
Using family type 'VSP Series' for this script

Input Data:
- Selected Switch IP = 10.8.4.5
- Input Config Template file = /root/Ludo/template.cfg
- Input CSV file = /root/Ludo/variable-values.csv
- Key to CSV data = IP address

Retained device 10.8.4.5 Site Variables:
{
  "Auto Sync VLANs in progress": "Done",
  "__PATH__": "/World/PoC/Sandbox ZTF",
  "core1": "",
  "core2": "",
  "dataIsid": "",
  "dataVlan": "",
  "deviceIP": "10.8.4.5",
```

Close

Show Output

Start Date/Time	End Date/Time	Message
7/18/2022 2:11:04 ...	7/18/2022 2:11:04 ...	Applied Co...
7/18/2022 2:11:04 ...	7/18/2022 2:11:04 ...	Applied Co...
7/18/2022 2:11:04 ...	7/18/2022 2:11:04 ...	Applied Co...
7/18/2022 2:11:04 ...	7/18/2022 2:11:04 ...	Applied Co...

Workflow Event signal



Alarms Alarm Configuration Events Event Configuration							
All Type: Console View Export to CSV							
Date/Time	Source	Subcomponent	Client	User	Type	Event	Information
1/18/2022 3:07:37 PM	[10.8.4.5, 10.8.4.3, 10....	Workflow: Apply Config Template	---	Istevens	Event	Configuration Template /root/Ludo/template.cfg applied	Configuration Template /root/Ludo/template.cfg applied to device [10.8.4.5, 10.8.4.3, 10.8.4.4, 10.8.4.2]

